A/B Testing Analysis Project

Case Study: Croatian E-Commerce Platform

Ahmad Miftahul Farohi
Mentor: Mas Hilmi

February 16, 2026

Contents

1 Introduction: The Business Problem

1.1 Why A/B Testing Matters in E-Commerce

E-commerce companies face a fundamental challenge: How do we know if changing our website will increase revenue?

Traditional approaches often rely on:

- HiPPO (Highest Paid Person's Opinion)
- Gut feelings (I think users will like this)
- Observational analysis, such as comparing before and after metrics which are often confounded by seasonality, trends, and external events

All of these approaches fail to establish causality. A/B testing provides a scientific and data-driven alternative.

What is A/B Testing?

A/B testing, also known as randomized controlled trials or controlled experiments, is the gold standard for measuring causal effects in digital products. The process consists of:

1. Randomly assigning users into:
   - Control (A): current experience
   - Treatment (B): new experience with exactly one change
2. Measuring key business metrics such as revenue, conversion rate, and engagement
3. Comparing the two groups to estimate the causal effect of the change

The key principle of A/B testing is randomization. Randomization ensures that both groups are statistically identical except for the applied change, allowing observed differences to be attributed solely to the treatment.

1.2 Our Case Study: Croatian E-Commerce Platform

Business Problem:

An online retailer aimed to test whether simplifying their checkout process would increase conversion rates without negatively affecting average order value. This case study demonstrates the complete A/B testing workflow, covering hypothesis formulation, validation checks, statistical analysis, and business decision-making.

Study Context:

- Platform: Major Croatian e-commerce company
- Timeline: March–June 2021 (Q1–Q2)
- Sample size: More than 102,000 user sessions
- Geography: Zagreb, Split, Rijeka, Osijek
- Devices:
  - Mobile (60%)
  - Desktop (35%)
  - Tablet (5%)

In this context, five different experimental tests were conducted to collect data in accordance with the study requirements. The five experiments include:

- Menu Design
- Novelty Slider
- Product Recommendation Sliders
- Customer Reviews
- Search Engine

These five experimental tests will be used to help answer the key question:

"How can we determine whether changes to our website will lead to an increase in revenue?"

By systematically measuring the impact of each change, we can make data-driven decisions and ensure that any modifications contribute positively to overall business performance.

2 Experiment Design

This experiment design aims to measure the impact of various changes to website features on user behavior and potential revenue growth. The method used is A/B testing, where users are randomly assigned to different groups, and each group is exposed to a different version of a feature. This approach allows the performance differences between versions to be compared objectively based on data.

There are five main experiments being tested, namely:

- Menu Design
  - Experiment: A_horizontal_menu vs B_dropdown_menu
  - Sample size: 7,000 rows
- Novelty Slider
  - Experiment: A_manual_novelties vs B_personalized_novelties
  - Sample size: 16,000 rows
- Product Recommendation Sliders
  - Experiment:
    * A_selected_by_others_only
    * B_similar_products_top
    * C_selected_by_others_top
  - Sample size: 18,000 rows
- Customer Reviews
  - Experiment: A_no_featured_reviews vs B_featured_reviews
  - Sample size: 42,000 rows
- Search Engine
  - Experiment: A_hybris_search vs B_algolia_search
  - Sample size: 19,000 rows

3 Validation Check

Before proceeding to the analysis stage, the experiment must undergo a series of validation checks to ensure that the results are reliable and trustworthy. This validation is important because if the sample distribution is unbalanced, user characteristics differ significantly between groups, or the data is unstable over time, the statistical conclusions may be misleading. In this analysis, three validation tests were performed, namely:

3.1 Sampe Ratio Mismatch

What is Sample Ratio Mismatch (SRM)?

The Sample Ratio Mismatch (SRM) Test is used to verify whether the distribution of sample sizes across each variant matches the expected ratio. This test is important to ensure that the randomization process in the experiment was conducted correctly.

The method used is the Chi-Square Goodness of Fit Test, which compares the actual sample size (observed) with the expected sample size (expected). The test produces a chi-square statistic and a *p-value* as the result.

This test uses a threshold of 0.001. If the *p-value* < 0.001, it indicates that a Sample Ratio Mismatch has occurred, suggesting a potential issue in the sample allocation process. In this case, the experiment is considered invalid, and the results cannot be used for further analysis.

Conversely, if the *p-value* ≥ 0.001, the sample distribution is considered consistent with the expected ratio, and the experiment can proceed to the next stage of analysis.

Result of SRM

Test	Sample Size	Variants	SRM Result
Test 1: Menu Design	7000	A_horizontal_menu: 3,500 (50%) B_dropdown_menu: 3,500 (50%)	PASS ($p = 1.000$)
Test 2: Novelty Slider	16000	A_manual_novelties: 8,000 (50%) B_personalized_novelties: 8,000 (50%)	PASS ($p = 1.000$)
Test 3: Product Sliders	18000	A_selected_by_others_only: 6,000 (33.3%) B_similar_products_top: 6,000 (33.3%) C_selected_by_others_top: 6,000 (33.3%)	PASS ($p = 1.000$)
Test 4: Customer Reviews	42000	A_no_featured_reviews: 21,000 (50%) B_featured_reviews: 21,000 (50%)	PASS ($p = 1.000$)
Test 5: Search Engine	19000	A_hybris_search: 9,500 (50%) B_algolia_search: 9,500 (50%)	PASS ($p = 1.000$)

All five experiments have balanced sample distributions in accordance with the planned randomization (with exact proportions of $50:50$ or $33.3:33.3:33.3$). The SRM test results show a PASS with a $p - value = 1.0000$ for each experiment, indicating that no mismatch was detected. This means that the user allocation across variants is valid, and statistical analysis can be continued.

3.2 Covariate Balance Verification

What is Covariate Balance Verification?

The Covariate Balance Check is used to ensure that the baseline characteristics (covariates) are evenly distributed across variants in the experiment. This test uses the Standardized Mean Difference (SMD) to measure the difference between variants.

For numerical covariates, the SMD is calculated based on the difference in means normalized by the pooled standard deviation. For categorical covariates, the SMD is calculated based on the difference in proportions of each category between variants.

This test uses a threshold of 0.2. If the SMD > 0.2, the covariate is considered imbalanced, indicating a potential bias in the sample allocation. Conversely, if the SMD ≤ 0.2, the covariate is considered balanced.

The maximum SMD value (max SMD) is used as the primary indicator to assess the overall balance between variants. The smaller the SMD value, the better the covariate balance, and the more valid the experiment is for further analysis.

Result of Covariate Balance

Test	Sample Size	SMD Value	Balance Status
Test 1: Menu Design	7000	0.026	OK (Balanced)
Test 2: Novelty Slider	16000	0.028	OK (Balanced)
Test 3: Product Sliders	18000	0.039	OK (Balanced)
Test 4: Customer Reviews	42000	0.014	OK (Balanced)
Test 5: Search Engine	19000	0.032	OK (Balanced)

All five experiments show very small Standardized Mean Difference (SMD) values, ranging from 0.014 to 0.039, which are well below the established threshold of 0.2. This indicates that the baseline user characteristics (covariates) are evenly distributed across all experiment variants.

3.3 Temporal Stability Check

What is Temporal Stability Check

The Temporal Stability Check is used to verify whether the sample size distribution for each variant remains stable over time. This test uses the Coefficient of Variation (CV), which is the ratio of the standard deviation to the mean of the daily sample size for each variant.

The threshold used is 0.2. If the CV value < 0.2, the sample distribution is considered stable, indicating that the variant allocation process was consistent throughout the experiment period. Conversely, if the CV value ≥ 0.2, the distribution is considered unstable and may indicate potential issues such as system disruptions, traffic fluctuations, or allocation bias.

The maximum CV value (max CV) is used as the primary indicator to assess overall stability. The smaller the CV value, the more stable the sample distribution, and the more valid the experiment is for further analysis.

Result of Temporal Stability Check

Test	Sample Size	CV Value	Temporal Status
Test 1: Menu Design	7000	0.057	OK (Stable)
Test 2: Novelty Slider	16000	0.038	OK (Stable)
Test 3: Product Sliders	18000	0.049	OK (Stable)
Test 4: Customer Reviews	42000	0.047	OK (Stable)
Test 5: Search Engine	19000	0.026	OK (Stable)

All experiments show low Coefficient of Variation (CV) values, ranging from 0.026 to 0.057, which are significantly below the established threshold of 0.2. These low CV values indicate that the number of samples allocated to each variant remained relatively consistent from day to day, without significant fluctuations.

4 Statistical Analysis

4.1 Choosing the Right Test

Every metric in a dataset requires a different statistical test. The choice depends on:

1. **Metric type**: Binary, continuous, count, or ordinal
2. **Data distribution**: Normal, skewed, heavy-tailed
3. **Sample size**: Large (CLT applies) vs small
4. **Variance equality**: Equal vs unequal variances

Decision Tree for Test Selection

1. **Step 1: Identify metric type**
   - **Binary (0/1, yes/no)**: Conversion, click, bounce → Proportion test
   - **Continuous (real numbers)**: Revenue, time on site, cart value → t-test or Mann-Whitney
   - **Count (non-negative integers)**: Page views, items purchased → Poisson test or Mann-Whitney
   - **Categorical (multiple categories)**: Product category, exit page → Chi-square test
2. **Step 2: Check distributional assumptions**
   - If continuous and approximately normal: Use t-test (parametric)
   - If continuous and heavily skewed: Use Mann-Whitney U (non-parametric)
   - If count data: Consider distribution (Poisson vs negative binomial)
3. **Step 3: Select speci c test**
   - Two variants: Two-sample test
   - >2 variants: ANOVA or Kruskal-Wallis

4.2 Calculate Sample Size

Sample size calculation is the process of determining the minimum number of samples required for each variant (control and treatment) so that the experiment has sufficient statistical power to detect a meaningful difference.

The purpose of this calculation is to:

- Avoid underpowered experiments with too small a sample size
- Ensure that experimental results are statistically reliable
- Reduce the risk of false negatives (failing to detect an effect that actually exists)

Parameters used:

- Baseline conversion rate: 5%
- Minimum Detectable Effect (MDE): 10%

- Significance level (α): 0.05
- Statistical power: 80%

4.3 Normality Check

Normality check is used to determine whether the data follow a normal distribution. The method used for this normality test is the Shapiro-Wilk test.

Hypotheses:

- H_0: the data are normally distributed
- H_1: the data are not normally distributed

The results of the normality check are used to identify and determine the appropriate statistical test method.

- If normal → use t-test (parametric)
- If not normal → use Mann-Whitney U test (non-parametric)

4.4 Hypothesis Testing

Hypothesis testing is used to determine whether there is a significant difference between the control and treatment groups.

4.4.1 Proportion Test

Proportion test is used to compare the proportions between two groups. It is intended to determine whether the treatment significantly increases the conversion rate.

It is used for binary metrics, such as:

- Conversion (0 or 1)
- Click or no click
- Purchase or no purchase

4.4.2 Mann-Whitney U Test

The Mann-Whitney U test is a non-parametric test used to compare two groups without assuming a normal distribution. This test is applied when the data are not normally distributed, to determine whether there is a significant difference between the control and treatment groups.

4.4.3 Welch's t-test (Comparison Test)

Welch's t-test is a version of the t-test that does not assume equal variances. This test is used as a comparison to ensure the consistency of results.

4.5 Multiple Testing Correction

Multiple Testing Correction is a technique used to adjust p-values when performing multiple statistical tests simultaneously. This is important because the more tests conducted, the higher the likelihood of false positives or incorrect conclusions.

One popular method commonly used is the Holm-Bonferroni correction. This method aims to control the Family-Wise Error Rate (FWER), which is the probability of obtaining at least one false positive across the entire set of tests. By applying this correction, we can ensure more valid test results and reduce the risk of misinterpretation due to multiple testing.

4.6 Relative Lift

Relative lift measures the percentage increase of the treatment group compared to the control group. This test aims to show the practical business impact, not just statistical significance.

Formula:

$$\text{Relative Lift} = \frac{\text{treatment} - \text{control}}{\text{control}} \times 100\%$$

4.7 Result

Calculate Sample Size

Required sample size per variant: 31,231

Normality Check

The result of normality check for continuous metric (Revenue per User) is:

- Control : Non-normal
- Treatment : Non-normal

Since both groups are not normally distributed, the Mann-Whitney U test is used as the primary method.

Proportion Test

Metric	$p - value$	Significant	Relative Lift
Conversation Rate	$9.397606e - 02$	False	21.317829

From the table, it can be seen that the treatment shows an increase in conversion rate, but it is not statistically significant. This means that the increase is not strong enough to conclude that the treatment truly improves the conversion rate.

Mann-Whitney U Test

Metric	$p - value$	Significant
Revenue per User	$9.554408e - 08$	True

From the table, it can be seen that the treatment results in a statistically significant increase in revenue. This indicates that the treatment has a real impact on revenue per user."

Welch's t-test (Comparison Test)

Metric	$p - value$	Significant	Relative Lift
Revenue per User	$2.501073e - 06$	True	9.680834

The results are consistent with those of the Mann-Whitney test, thereby increasing confidence in the experimental findings.

5 Interpretation on Case Study

5.1 Test 1: Menu Design

Expreiment Details:

- **Sample Size:** 7,000 users (3,500 per variant)
- **Variants:**
  - A: Horizontal navigation menu (control)
  - B: Horizontal dropdown menu (treatment)

Validation Results:

- SRM Test: $p = 1.000$ ✓(perfect 50/50 split)
- Covariate Balance: Good balance (max SMD < 0.1) ✓

Statistical Results:

Metric	Horizontal (A)	Dropdown (B)	Lift	$P - Value$
Revenue (median)	€2.86	€2.60	-9.1%	< 0.001
Added to Cart	96.2%	86.2%	-10.4%	< 0.001
Pages Viewed (median)	2.17	2.13	-1.8%	0.202
Bounced	43.4%	44.5%	+2.5%	0.671

Statistical Conclusion:

- **Significant negative effects** on revenue and add-to-cart rate
- Dropdown menu *harms* key business metrics
- Effects remain significant after Holm-Bonferroni correction
- Data non-normal; Mann-Whitney U test used

Business Recommendation: DO NOT DEPLOY dropdown menu

- **Keep horizontal menu** - control outperforms treatment
- Dropdown menu reduces revenue by 9% and add-to-cart by 10%
- This is a *negative result* treatment made things worse

5.2 Test 2: Novelty Slider

Expreiment Details:

- **Sample Size:** 16,000 users (8,000 per variant)
- **Variants:**
  - A: Manually curated novelties (control)
  - B: Algorithm-personalized novelties (treatment)

Validation Results:

- SRM Test: $p = 1.000$ ✓(perfect 50/50 split)
- Covariate Balance: Excellent ✓

Statistical Results:

Metric	**Manual (A)**	**Personalized (B)**	**Lift**	$P - Value$
Novelty Revenue (median)	€3.77	€3.98	+5.6%	< 0.001
Products Added (median)	0	0	-	< 0.001
Is Registered	45%	45%	0%	1.000

Statistical Conclusion:

- **Significant improvement** in novelty revenue
- Products added shows distributional difference (both medians = 0, but significant p-value)
- Effects robust to multiple testing correction

Business Recommendation: DEPLOY PERSONALIZED NOVELTY SLIDER

- Clear winner with 5.6% revenue increase
- Personalization algorithm outperforms manual curation

5.3 Test 3: Product Recommendation Sliders

Expreiment Details:

- **Sample Size:** 18,000 users (6,000 per variant)
- **Variants:**
  - A: "Selected by others" only (social proof)
  - B: "Similar products" (content-based)
  - C: Hybrid (social + algorithmic)

Validation Results:

- SRM Test: $p = 1.000$ ✓(perfect 3-way split)
- Covariate Balance: Excellent ✓

Statistical Results:

Metric	Social (A)	Similar (B)	Lift	$P - Value$
Revenue from Rec. (median)	€3.72	€4.50	+21.0%	< 0.001
Products per Order (median)	3.16	3.05	-3.5%	< 0.001
Avg Product Price (median)	€3.01	€3.42	+13.6%	< 0.001
Add-to-Cart Rate (median)	0	0	-	1.000
Slider Interactions (median)	2	2	-	0.161

Statistical Conclusion:

- **Variant B signi cantly outperforms A** on revenue (+21%)
- Fewer products per order but higher-priced items (+13.6%)
- Strategy: Upselling to premium products drives revenue

Business Recommendation: DEPLOY VARIANT B (SIMILAR PRODUCTS)

- Highest revenue impact: +21% from recommendations
- Content-based algorithm superior to social proof alone
- Mechanism: Promotes higher-value products e ectively

5.4 Test 4: Customer Reviews

Expreiment Details:

- **Sample Size:** 42,000 users (21,000 per variant)
- **Variants:**
  - A: No featured reviews (control)
  - B: Featured reviews highlighted (treatment)

Validation Results:

- SRM Test: $p = 1.000$ ✓(perfect 50/50 split)
- Covariate Balance: Excellent ✓

Statistical Results:

Metric	No Reviews (A)	Featured (B)	Lift	$P - Value$
Conversion Rate	-	-	-	1.000
Added to Cart	-	-	-	< 0.933

Statistical Conclusion:

- **No significant effects detected** on any metric
- Large sample size (42,000) provides adequate power
- This is a valid *null result*

Business Recommendation: DO NOT DEPLOY featured reviews

- No evidence of improvement
- Focus resources on proven features

5.5 Test 5: Search Engine

Expreiment Details:

- **Sample Size:** 19,000 users (9,500 per variant)
- **Variants:**
  - A: SAP Hybris search (control)
  - B: Algolia search (treatment)

Validation Results:

- SRM Test: $p = 1.000$ ✓(perfect 50/50 split)
- Covariate Balance: Excellent ✓

Statistical Results:

Metric	Horizontal (A)	Dropdown (B)	Lift	$P - Value$
Avg Revenue (median)	€0.69	€0.69	0%	1.000
Added to Cart	89.7%	91.6%	+2.1%	< 0.010
Conversion Rate	-	-	-	1.000
Search Interaction	-	-	-	1.000

Statistical Conclusion:

- **Significant improvement** in add-to-cart (+2.1%)
- No signi cant e ect on revenue or conversion
- Modest but measurable engagement improvement

Business Recommendation: CONDITIONAL DEPLOYMENT

- Small improvement vs. subscription cost
- Deploy if already planning infrastructure upgrade
- Otherwise, optimize current Hybris search first

5.6 Summary of All 5 Experiments

Test	N	Result	Key Effect	Decision
Menu	7K	Negative	-9% revenue	DON'T DEPLOY
Novelty	16K	Positive	+5.6% revenue	DEPLOY
Sliders	18K	Positive	+21% revenue	DEPLOY (B)
Reviews	42K	Null	No effect	DON'T DEPLOY
Search	19K	Marginal	+2.1% cart	Conditional

Overall Success Rate:

- **Clear Positive:** 2/5 (Tests 2, 3) 40%
- **Negative:** 1/5 (Test 1) 20%
- **Positive:** 1/5 (Test 4) 20%
- **Marginal:** 1/5 (Test 5) 20%

Key Learnings:

- Not all hypotheses succeed this is scientifically normal
- Negative results prevent harmful deployments
- Null results inform resource allocation
- Largest effects from algorithmic improvements (Test 3: +21%)
- All validation checks passed high confidence in results

6 Business Recommendation & ROI Analysis

6.1 Prioritized Deployment Roadmap

Based on impact, con dence, and implementation complexity:

1. **Immediate (Week 1-2:**
   - **Deploy Test 5:** Algolia search (highest ROI, low risk)
   - **Deploy Test 2:** Personalized novelty slider (large effect, medium complexity)
2. **Short-term (Week 36):**
   - **Deploy Test 4:** Featured reviews (clear benefit, easy implementation)
   - **Deploy Test 3:** Hybrid product recommendation (moderate complexity)
3. **Do Not Deploy:**
   - **Test 1:** Dropdown menu (no benefit, negative trend)

6.2 Lessons from Each Test

Test 1 (Menu) Not all changes improve metrics. Negative trends (even if not significant) warrant caution.

Test 2 (Novelty) Personalization can have massive impact. Algorithm outperformed human curation by 20%+.

Test 3 (Recommendations) Multiple variant tests (ANOVA) provide richer insights than simple A/B.

Test 4 (Reviews) CUPED variance reduction enabled tighter CIs and better ROI estimates.

Test 5 (Search) Third-party tools can dramatically outperform native solutions. ROI analysis justifies cost.